

Topology PhD Exam, August 2025

Be neat and do not write too small.

For the first five problems, show all of your work and support all statements.

1. Let (X, d) be a compact metric space. Let $f : X \rightarrow X$ and suppose there is some $c < 1$ with $d(f(x), f(y)) \leq c \cdot d(x, y)$ for all $x, y \in X$. Prove there exists a unique $x \in X$ with $f(x) = x$.
2. Pick a CW complex structure on the Klein bottle K , and then use cellular homology to compute $H_i(K; \mathbb{Z})$ for $i = 0, 1, 2$.
3. This problem has two parts.
 - (a) Show that the antipodal map $-1 : S^n \rightarrow S^n$ defined by $-1(x) = -x$ has degree $(-1)^{n+1}$.
 - (b) Prove that S^n has a continuous nonzero tangent vector field if and only if n is odd.
4. Let M be a compact orientable n -manifold without boundary. Prove that if n is odd, then the Euler characteristic $\chi(M)$ is zero.
5. Let M_1 and M_2 be closed connected n -manifolds, and let $M_1 \# M_2$ be their connected sum obtained by removing a ball from each and gluing along the resulting boundary spheres. Show that there are isomorphisms $H_i(M_1 \# M_2; \mathbb{Z}) \cong H_i(M_1; \mathbb{Z}) \oplus H_i(M_2; \mathbb{Z})$ for $0 < i < n$, with one exception: If both M_1 and M_2 are nonorientable, then $H_{n-1}(M_1 \# M_2; \mathbb{Z})$ is obtained from $H_{n-1}(M_1; \mathbb{Z}) \oplus H_{n-1}(M_2; \mathbb{Z})$ by replacing one of the two $\mathbb{Z}_2$ summands by a $\mathbb{Z}$ summand.

Answer the following ten problems with complete definitions, complete statements, an example, or a short proof.

6. Show that if $g : S^2 \rightarrow S^2$ is continuous and $g(x) \neq g(-x)$ for all $x \in S^2$, then g is surjective.
7. State the Tychonoff Theorem.
8. Show that $\mathbb{R}^\omega$ with the box topology is not connected.
9. Let N_g be the nonorientable surface of genus g , i.e., the g -fold connected sum of projective planes. Use the Seifert–van Kampen theorem to derive the fundamental group of N_g for $g \geq 1$.
10. Let X be two unlinked circles in $\mathbb{R}^3$ and let Y be two simply linked circles. Show that the complements $\mathbb{R}^3 \setminus X$ and $\mathbb{R}^3 \setminus Y$ are not homotopy equivalent.
11. Show that if n is even, then every map $f : \mathbb{RP}^n \rightarrow \mathbb{RP}^n$ has a fixed point.
12. Let $p_1 : (\tilde{X}_1, \tilde{x}_1) \rightarrow (X, x_0)$ and $p_2 : (\tilde{X}_2, \tilde{x}_2) \rightarrow (X, x_0)$ be covering space maps, where all of the involved spaces are path-connected and locally path-connected. An isomorphism of covering spaces $f : (\tilde{X}_1, \tilde{x}_1) \rightarrow (\tilde{X}_2, \tilde{x}_2)$ is a homeomorphism satisfying $p_1 = p_2 f$. Prove the covering spaces p_1 and p_2 are isomorphic if and only if $p_{1*}(\pi_1(\tilde{X}_1, \tilde{x}_1)) = p_{2*}(\pi_1(\tilde{X}_2, \tilde{x}_2))$.
13. Define what it means for a subspace $A \subseteq X$ to be a retract. Give an example of a retract $A \subseteq X$ that is not a deformation retract.

14. Use the Tietze extension theorem to prove Urysohn's lemma.
15. What are the cohomology groups and cup product structure in $H^*((S^2 \times S^8) \# (S^4 \times S^6); \mathbb{Z})$, where $\#$ denotes the connected sum of the manifolds? You do not need to prove your answer.